

A Survey on Unit Testing Practices and Problems

Publisher: **IEEE** [Cite This](#)  PDF

Ermira Daka ; Gordon Fraser **All Authors**

76
Cites in
Papers

5138
Full
Text Views



Abstract
Document Sections
I. Introduction
II. Survey Design
III. Survey Results
IV. Related Work
V. Conclusions

Authors
Figures
References
Citations
Keywords
Metrics
More Like This
Footnotes

Abstract:

Unit testing is a common practice where developers write test cases together with regular code. Automation frameworks such as JUnit for Java have popularised this approach, allowing frequent and automatic execution of unit test suites. Despite the appraisals of unit testing in practice, software engineering researchers see potential for improvement and investigate advanced techniques such as automated unit test generation. To align such research with the needs of practitioners, we conducted a survey amongst 225 software developers, covering different programming languages and 29 countries, using a global online marketing research platform. The survey responses confirm that unit testing is an important factor in software development, and suggest that there is indeed potential and need for research on automation of unit testing. The results help us to identify areas of importance on which further research will be necessary (e.g., Maintenance of unit tests), and also provide insights into the suitability of online marketing research platforms for software engineering surveys.

Published in: 2014 IEEE 25th International Symposium on Software Reliability Engineering

Date of Conference: 03-06 November 2014 **DOI:** 10.1109/ISSRE.2014.11

Date Added to IEEE Xplore: 15 December 2014 **Publisher:** IEEE

► ISBN Information: **Conference Location:** Naples, Italy

▼ ISSN Information:

Sign in to Continue Reading

Authors	▼
Figures	▼
References	▼
Citations	▼
Keywords	▼
Metrics	▼
Footnotes	▼



Need
Full-Text

access to IEEE *Xplore*
for your organization?

CONTACT IEEE TO SUBSCRIBE >

IEEE Personal Account

CHANGE
USERNAME/PASSWORD

Purchase Details

PAYMENT OPTIONS
VIEW PURCHASED
DOCUMENTS

Profile Information

COMMUNICATIONS
PREFERENCES
PROFESSION AND
EDUCATION
TECHNICAL INTERESTS

Need Help?

US & CANADA: +1 800
678 4333
WORLDWIDE: +1 732
981 0060
CONTACT & SUPPORT

Follow

[f](#) [@](#) [in](#) [▶](#)

About IEEE *Xplore* | Contact Us | Help | Accessibility | Terms of Use | Nondiscrimination Policy | IEEE Ethics Reporting [🔗](#) | Sitemap | IEEE Privacy Policy

A public charity, IEEE is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity.

© Copyright 2025 IEEE - All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies.